THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 9: **Propositions and their Converses** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** If a number is divisible by 5, then it is divisible by 10.	swap the 'if' part and the 'then' part
2	**(b)** If a number is non-negative, then it is a perfect square.	swap the 'if' part and the 'then' part
3	**(a)** 7	it satisfies the 'if' part but not the 'then' part
4	**(a)** both the proposition and its converse are true	the square of an odd number is odd
5	**(d)** right-angled, since $a^2 + b^2 = c^2$	$400 + 441 = 841$ and $29^2 = 841$
6	**(c)** 3 factors; it is a perfect square	count the factor pairs; an odd count signals a repeated pair
7	**(b)** 3	$121 = 11^2$
8	**(c)** the same meaning	interchangeable forms
9	**(a)** both true, or both false, or one true and one false	all four combinations can occur in general
10	**(c)** If the angles opposite two sides of a triangle are equal, the two sides are equal.	the base-angle theorem and its converse

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Converse: If the alternate interior angles formed by a transversal are equal, then the two lines are parallel (1). It is true (1).	1+1
12	Converse: If a number is divisible by 9, then it is divisible by 18 (1). False: 27 is divisible by 9 but not by 18 (1).	1+1
13	Converse: If $x^2 = y^2$, then $x = y$ (1). Counterexample: $x = 3,\ y = -3$ gives $x^2 = y^2$ but $x \neq y$ (1).	1+1
14	For $n = 11$: $121 + 11 + 11 = 143 = 11 \times 13$, which is composite (2).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Proposition: true, since $15 = 5 \times 3$ (1). Converse: If a number is divisible by 5, then it is divisible by 15 (1). False: 10 is divisible by 5 but not by 15 (1).	1+1+1
16	Converse: If the corresponding angles of two triangles are equal, then the triangles are congruent (1). False (1). Counterexample: two equilateral triangles with sides 3 cm and 5 cm have equal angles but are not congruent (1).	1+1+1
17	For $n = 11$: $n^2 + n + 11 = 121 + 11 + 11 = 143$ (1). $143 = 11 \times 13$ (1), so it is composite and the claim is false (1).	1+1+1
	OR: Largest side 37: $12^2 + 35^2 = 144 + 1225 = 1369$ (1). $37^2 = 1369$ (1). By the converse of the Baudhāyana–Pythagoras theorem the triangle is right-angled (1).	1+1+1
18	Justification: $6 = 3 \times 2$, so every multiple of 6 is a multiple of 3 (1). Converse: If a number is a multiple of 3, then it is a multiple of 6 (1). Counterexample: 9 (1).	1+1+1
	OR: Pairs: (1,36), (2,18), (3,12), (4,9), (6,6) (1½). The pair (6,6) repeats (½). The count is $2 \times 4 + 1 = 9$, odd, because 36 is a perfect square (1).	1½+½+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) If the road is wet, then it has rained (1). (ii) A water tanker spills water on the road, or a drain overflows (1). (iii) Yes, in normal situations rain makes roads wet (1). (iv) A counterexample is an instance that satisfies the 'if' part but not the 'then' part (1); one such case shows quickly that a proposition is false (1).	1+1+1+1+1
	OR: Construct right triangle XYZ with perpendicular sides $YZ = a$, $XZ = b$ (1). By the theorem $XY^2 = a^2 + b^2 = c^2$, so $XY = c$ (1). In triangles ABC and XYZ: $BC = YZ = a$, $AC = XZ = b$, $AB = XY = c$ (1). By SSS the triangles are congruent (1). So $\angle C = \angle Z = 90^\circ$ and $\triangle ABC$ is right-angled (1).	1+1+1+1+1

20	(i) 49: pairs (1,49), (7,7) → factors 1, 7, 49: 3 (odd) (1). 100: pairs (1,100), (2,50), (4,25), (5,20), (10,10) → 9 factors (odd) (1½). (ii) Converse: If a number has an odd number of factors, then it is a perfect square (1). True: an odd count forces a repeated pair (f, f), so $n = f \times f$ (1½).	2½+2½
	OR: (i) $1 + 1 + 11 = 13$, $4 + 2 + 11 = 17$, $9 + 3 + 11 = 23$, $16 + 4 + 11 = 31$ (1½). (ii) No — a few cases do not prove a general claim (1). (iii) $n = 10$: $100 + 10 + 11 = 121 = 11^2$ is not prime; (smaller n up to 9 give 13, 17, 23, 31, 41, 53, 67, 83, 101 — all prime) (1½). (iv) It is a counterexample (1).	1½+1+1½+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 9: **Propositions and their Converses** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** If two angles are equal, then they are vertically opposite.	swap the 'if' part and the 'then' part
2	**(d)** If $x^2 = 9$, then $x = 3$.	swap the 'if' part and the 'then' part
3	**(c)** 20	it satisfies the 'if' part but not the 'then' part
4	**(a)** $x = 3$ satisfies $x > 2$ but not $x > 5$	a counterexample for the converse
5	**(b)** not right-angled, since $a^2 + b^2 \neq c^2$	$81 + 144 = 225$ and $16^2 = 256$
6	**(d)** 9 factors; it is a perfect square	count the factor pairs; an odd count signals a repeated pair
7	**(c)** 4	$35 = 5 \times 7$
8	**(d)** If X then Y	another way to write the proposition
9	**(b)** "If a triangle is equilateral, then all its angles are equal" and "If all angles of a triangle are equal, then it is equilateral"	both directions hold
10	**(a)** Both A and R are true and R is the correct explanation of A	SSS congruence proves the converse

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	For $n = 11$: $2^{11} - 1 = 2047 = 23 \times 89$ is composite although 11 is prime (2).	1+1
12	9 is odd, so the 'if' part holds (1). $9 = 3 \times 3$ is not prime, so the 'then' part fails (1).	1+1
13	Converse: If $a^2 + b^2 = c^2$ in a triangle with sides a, b, c, then it is right-angled (1). It is true, by SSS congruence with a right triangle of legs a and b (1).	1+1
14	True (½). Converse: If the diagonals of a quadrilateral are equal, then it is a rectangle (½). Counterexample: an isosceles trapezium has equal diagonals but is not a rectangle (1).	½+½+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	All other factors form pairs of two different numbers, giving an even count (1). The repeated pair (f, f) contributes just one factor f (1). Even + 1 = odd (1).	1+1+1
16	Proposition: true, $a = m^2,\ b = n^2 \Rightarrow ab = (mn)^2$ (1). Converse: If ab is a perfect square, then a and b are perfect squares (1). False: $a = 2,\ b = 8$ gives $ab = 16$ but 2 is not a perfect square (1).	1+1+1
17	(i) $n = 5$ is prime but $n + 4 = 9 = 3 \times 3$ is not prime, so $n = 5$ is a counterexample (1½). (ii) $n = 10$ is a multiple of 5 but does not end in 5 (1½).	1½+1½
	OR: Converse: If n is divisible by both 5 and 12, then it is divisible by 60 (1). 5 and 12 have no common factor (1), so $5 \times 12 = 60$ divides n: the converse is true (1).	1+1+1
18	$n = 12$ is divisible by 4 and 6 but not by 24 (1½). 4 and 6 share the factor 2, so their product 24 is not the smallest common multiple; the LCM is 12 (1½).	1½+1½
	OR: Converse: If our school has an assembly, then it is Monday (1). The school may also hold assemblies on other days, e.g. on Independence Day (1). So an assembly day need not be Monday; the converse can be false (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Proposition true (½). Converse: If $x^3 = y^3$ then $x = y$ — true for real numbers because the cube is one-to-one (e.g. $x^3 = -8 \Rightarrow x = -2$) (2). (ii) Proposition true (½). Converse: If $x^2 = y^2$ then $x = y$ — false: $x = 2,\ y = -2$ (2).	2½+2½
	OR: $AB = 7$, $AC = 24$, $BC = \sqrt{49 + 576} = \sqrt{625} = 25$ (2). $AB^2 + AC^2 = 49 + 576 = 625 = BC^2$, so by the converse the triangle is right-angled at A (1½). Area $= \frac{1}{2} \times 7 \times 24 = 84$ square units (1½).	2+1½+1½

20	(i) The converse of the Baudhāyana–Pythagoras theorem (1). (ii) $3^2 + 4^2 = 9 + 16 = 25 = 5^2$ (1). (iii) The converse: we know the sides and want to conclude the right angle (1). (iv) Yes, $5^2 + 12^2 = 25 + 144 = 169 = 13^2$ (2).	1+1+1+2
	OR: (i) Factors of p^2 are $1, p, p^2$ only, because the only divisors of p^2 are products of the single prime p: 3 factors (2). (ii) Converse: If n has exactly 3 factors, then $n = p^2$ for a prime p (1). Three factors is odd, so n is a perfect square $f \times f$; if f had a proper factor g other than 1, then g would give a fourth factor beyond $1, f, f^2$; so f is prime (2).	2+3

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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